Unit-V

5. (*a*) The following table shows the jobs of a network along with their time estimates. The time estimates are in days :

Job	1-2	1-6	2-3	2-4	3-5	4-5	5-8	6-7	7-8
to	3	2	6	2	5	3	1	3	4
tm	6	5	12	5	11	6	4	9	19
tp	15	14	30	8	17	15	7	27	28

(i) Draw the project network.
(ii) Find the critical path and standard deviation.
(iii) Find the probability of the project being completed in 31 days.
[Given area 0 to 2.166 = 0.4886]. 8

(*b*) A small maintenance project consists of the following jobs, whose precedence relationships are given below :

Job	1-2	1-3	2-3	2-5	3-4	3-6	4-5	4-6	5-6	6-7
Duration (days)	15	15	3	5	8	12	1	14	3	14

(i) Draw an arrow diagram representing the project.
(ii) Find the critical path and the total project duration.
(iii) Calculate the total float for each activity. 8

(*c*) Discuss the similarities and differences of CPM and PERT. 8

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MBA (3rd Semester) Examination, Nov.-Dec., 2015

(New Scheme)

Optimization Methods

Time Allowed : 3 hours *Maximum Marks* : 80
Minimum Pass Marks : 32

Note : (*i*) Attempt any **two** parts from (*a*), (*b*) and (*c*) of each question. Use of calculator, log tables and statistical tables is permitted.
(*ii*) The figures in the right-hand margin indicate marks.

Unit-I

1. (*a*) A firm manufacturers two products A and B on which the profits earned per unit are Rs. 3 and Rs. 4 respectively. Each product is processed on two machines M1 and M2. Product A requires one minute of processing time on M1 and two minutes on M2; while B requires one minute on M1 and one minute on M2.

Machine M1 is available for not more than 7 hrs. 30 minutes; while machine M2 is available for 10 hr. during any working day. Find the number of units of products A and B to be manufactured to get maximum profit. 8

(*b*) Food X contains 6 units of vitamin A per gram and 7 units of vitamin B per gram and costs 12 paise per gram. Food Y contains 8 units of vitamin A per gram and 12 units of vitamin B per gram and costs 20 paise per gram. The daily minimum requirement of vitamin A and vitamin B is 100 units and 120 units respectively. Find the minimum cost of product mix by simplex method. 8

(*c*) What is the significance of the duality theory of linear programming? Describe the general rules for writing the dual of linear programming problem. 8

Unit-II

2. (*a*) Consider the following unbalanced transportation problem :

		To			Supply
From		D1	D2	D3	
	O1	5	1	7	10
	O2	6	4	6	80
	O3	3	2	5	50
Demand		75	20	50	

Since there is not enough supply, some of the demands at these destinations may not be satisfied. Suppose there are penalty costs

for every unsatisfied demand unit which are given by 5, 3 and 2 for destination D1, D2, and D3 respectively. Find the optimal solution. 8

(*b*) There are three factories A, B and C, which supply goods to four dealers D1, D2, D3, D4. The production capacities of these factories are 1000, 700 and 900 units per month respectively. The requirements from the dealers are 900, 800, 500 and 400 units per month respectively. The per unit returns (excluding transportation cost) are Rs. 8, Rs. 7 and Rs. 9 at three factories. The following table gives are unit transportation costs from the factories to the dealers :

	D1	D2	D3	D4
A	2	2	2	4
B	3	5	3	2
C	4	3	2	1

8

Determine the optimum solution to maximize the total returns.

(*c*) Write a note on tracing a closed loop. What are the characteristics features of a closed loop? 8

Unit-III

3. (*a*) A firm produces four products. There are four operators who are capable of producing each of them. The processing time varies from operator to operator. The firm works 8 hours a day and allows 30 minutes for lunch.

The processing times in minutes and profits of each of the product are given below :

Operators	Products			
	A	B	C	D
1	15	9	10	6
2	10	6	9	6
3	25	15	15	9
4	15	9	10	10
Profit (Rs/unit)	8	6	5	4

Find the optimal assignment of each product to operators. 8

(*b*) Five wagons W1, W2, W3, W4 and W5 are available at stations. These are required at five stations I, II, III, IV, and V. The mileages between various stations are given by the table below. How should be wagons be transported so as to minimize the total mileage covered? 8

	I	II	III	IV	V
W1	10	5	9	18	11
W2	13	9	6	12	14
W3	3	2	4	4	5
W4	18	9	12	17	15
W5	11	6	14	19	10

(*c*) What is the travelling salesman model? What are its practical applications? How does its solution differ from the solution of the assignment problem? 8

Unit-IV

4. (*a*) You are given the following data regarding the processing times (in hr.) of some jobs on three machines MI, MII and MIII. The order of processing is MI-MII-MIII. Determine the sequence that minimizes the total elapsed time (T) required to comple the jobs. Also evaluate T and idle time of MII and MIII. : 8

Jobs	A	B	C	D	E	F	G
MI	3	8	7	4	9	8	7
MII	4	3	2	5	1	4	3
MIII	6	7	5	11	5	6	12

(*b*) A TV Repair man finds that the time spent on his job has an exponential distribution with mean 30 minutes, if he repairs sets in order in which they come, and if the arrival of sets is approximately Poisson with an average rate of 10 per 8 hours day. What is the repair man's expected idle time each day? How many jobs are ahead of the set just brought in? 8

(*c*) What is a queuing problem? Explain the general structure of the queuing system? 8